AI-Orchestrated Diagnosis and Repair of a VQE Ansatz Failure Mode

A real, reproducible worked example of using an AI agent (Claude, via Claude Code) to orchestrate a multi-week, compute-heavy quantum chemistry research workflow — submitted as the optional prototype for Track 1 (AI for Quantum Advantage on NISQ Hardware). Code and full results: `https://github.com/1shaanSr/vqe-ansatz-expressibility`

What this document is

This is not a description of how AI agents *could* be used for quantum research — it is an account of a specific project where one was, with the exact commands, failures, self-corrections, and validated results it produced. The underlying scientific result (Section 3) and the AI-orchestration process that produced it (Section 2) are both real and independently verifiable from the linked repository's commit history, run logs, and figures.

1. The research question

We study a chemistry-structured VQE ansatz (paired UCC doubles, PUCCD) across the potential energy surfaces of five molecules — H_2, LiH, BeH_2, HF, BH (4–12 qubits, STO-3G) — asking two questions: where does this ansatz fail, and is a given failure a fundamental limitation of what the ansatz can represent, or just an under-optimized run? The second question is rarely asked rigorously in small-scale VQE studies, and answering it is where AI-agent orchestration did the most work.

2. Why this fits Track 1

Everything in this project to date, including the ansatz fix in Section 3, has been validated only in noiseless classical simulation; no run has yet been submitted to real quantum hardware. The proposed extension (Section 4) is designed around a single-batched-job execution strategy chosen specifically to stay within free-tier QPU-time limits (all classical optimization performed in simulation; hardware used only for final verification) — a direct, concrete push toward results that are harder to reach with today's NISQ devices under realistic access constraints.

2. The AI-orchestration workflow (the "how")

Three concrete episodes from this project illustrate what "AI-orchestrated" meant in practice — not passively running pre-written scripts, but actively managing long, failure-prone compute jobs and catching problems a human running the same pipeline could easily have missed.

2.1 Long-running background job orchestration

The full study spans 87 bond-length geometries across 5 molecules, each requiring a multi-restart classical optimization (up to 12 restarts × 250 iterations per geometry). Individual runs took minutes to hours depending on system load, with highly variable per-point timing (observed range: ~150 seconds to over 2 hours for a single geometry under heavy system contention). The agent launched these as background processes, monitored them asynchronously without blocking on results, diagnosed and fixed real runtime bugs as they appeared (e.g. a silent NaN-target crash from 4 incomplete database records, a matplotlib API incompatibility), and re-launched corrected runs — all without requiring a human to babysit each step.

2.2 Catching its own overstated statistical claim

An early analysis reported a strong correlation between energy error and dipole-moment error (naive $p = 3.4\times10^{-5}$). Before accepting this, the agent identified that the underlying data points are *not* independent samples — they lie along one smooth curve — and applied an autocorrelation-corrected effective-sample-size test (Pyper & Peterman, 1998), which revised the result to a far less dramatic $p \approx 0.035$. A second self-audit, using outlier-robust Spearman rank correlation, further revealed that for two of the five molecules, an apparently strong raw correlation ($r = 0.75$–0.91) was being driven entirely by 1–2 extreme data points, not a real trend — and reported this honestly rather than keeping the more impressive-looking number.

2.3 Refusing to accept a good-looking result at face value

After a proposed ansatz fix showed a dramatic improvement (up to 446× error reduction) at 4 hand-picked, previously-catastrophic geometries, the agent did not stop there. It proactively re-ran the fix across the *full* set of 55 geometries for the affected molecules — precisely because a 4-point spot check could have been cherry-picked or masked a regression elsewhere on the curve. The full-curve run (Section 3) confirmed the fix generalized, and also surfaced an honest partial result (one molecule improved substantially but not completely) that a less rigorous process might have glossed over.

3. The scientific result this process produced (the "what")

The orchestration process above was in service of a real finding, not just activity for its own sake. Two distinct ansatz failure modes were identified and mechanistically explained, then one was resolved:

3.1 Two failure modes, explained by a classical (non-VQE) diagnostic

LiH's energy and dipole error degrade *gradually* as the bond stretches. HF and BH instead track the exact answer closely, then collapse sharply at one specific bond length. A purely classical diagnostic — the weight of the Hartree-Fock determinant in the exact (FCI) wavefunction, requiring no VQE run at all — explains why: HF and BH each undergo a genuine electronic-structure curve crossing at essentially the bond length where PUCCD fails; LiH does not.

3.2 A validated architectural fix

PUCCD is restricted to paired double excitations and has no orbital-rotation degree of freedom — exactly the capability needed to track a wavefunction across a crossing. This is grounded directly in Anselmetti et al. (*New J. Phys.* 23, 113010, 2021), who show that pair-exchange-only and orbital-rotation-only ansatze are each individually non-universal, while their combination is. Adding single (orbital-rotation) excitations to PUCCD was tested across all affected geometries, not just the original failure points:

Molecule	Geometries tested	Reached chemical accuracy	Max error reduction
HF	17	16 / 17 (was 0 / 17)	~38,766x
LiH	20	20 / 20 (was 0 / 20)	~28,943x
BH	18	1 / 18, every point improved	446x

Chemical accuracy threshold: 1.6 mHa. BH's partial result (reported honestly, not rounded up) is itself informative — it suggests boron's valence configuration retains residual multi-reference character that orbital rotation alone does not fully capture, a concrete lead for the proposed extension below.

3.3 Ruling out the obvious alternative explanation

Before attributing the original failures to the ansatz, insufficient optimization was explicitly ruled out: at HF's worst-affected geometries, 12 independent restarts (random and warm-started) converged to the identical energy to within 1×10^{-6} Ha regardless of starting point — a genuine expressibility ceiling, not an under-optimized run. BH and LiH show a rougher multi-restart landscape at a subset of geometries (spreads up to roughly 1 Ha near BH's curve crossing); all reported energies use the best of twelve restarts to control for this.

4. Proposed extension using Quantum Flywheel resources

The fix above is validated only in noiseless simulation. The proposed 3-month extension uses exactly the AI-orchestration pattern demonstrated in Section 2, now applied with real QPU access, more compute, and AI-agent-driven search:

- **Month 1 — QPU access.** Validate the fix on real IBM hardware at the 55 decisive geometries (17 HF, 20 LiH, 18 BH), evaluating already-converged simulation parameters once per geometry (no on-hardware re-optimization). The measurement cost is computed, not guessed: each molecule's Jordan-Wigner-mapped 10-qubit Hamiltonian (276 Pauli terms) groups into 58 qubit-wise-commuting circuits, giving a bottom-up budget of ~6.4×10^6–1.3×10^7 shots at 2,000–4,000 shots/circuit. *Success criterion:* hardware energies agree with simulation to within chemical accuracy (1.6 mHa) at a majority of geometries, with any systematic deviation characterized and reported.
- **Month 2 — cloud compute.** Test generalization two ways: three new molecules with qualitatively different bonding (H_2O, NH_3, CO) in the same tractable qubit range, and one existing molecule re-run in an extended basis (6-31G vs. STO-3G). In parallel, target BH's residual gap (Section 3.2) with a small feedforward neural correction (1–2 hidden layers, $O(10^2)$ parameters) in the style of Li et al., trained on BH's existing 18-geometry dataset. *Success criterion:* the fix's benefit replicates on at least one new molecule/basis, and the correction narrows BH's gap beyond its current 446× best case.
- **Month 3 — AI-agent-orchestrated search.** Rather than hand-selecting excitation operators, use AI-agent-driven search over the untried space — generalized (non-paired) doubles and higher-order singles — targeted at BH's largest-residual geometries. Every candidate is run through the identical audit already validated on the original fix (matched-baseline comparison, twelve-restart consistency check, autocorrelation-corrected statistics from Section 2.2), so a candidate is reported as real only after surviving the same scrutiny the original result did.

5. Why this is a low-execution-risk proposal

This is not a cold-start proposal. The pipeline, the statistical methodology, and the diagnostic tooling all already exist and are already validated in simulation — only the hardware-execution step itself (Month 1) is new. The 3-month plan extends a demonstrated, working process rather than building one from scratch. All code, run scripts, raw results, and figures referenced in this document are public: `https://github.com/1shaanSr/vqe-ansatz-expressibility`.